BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: PR–IX (A-I-A4) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–IX (A-I-A4) (24)

Note: Note: Four circles are given in front of each question. Fill only the circle against the correct answer. Use a blue or black pen or marker to shade (fill) the circles. Filling more than one circle for a question will make that answer wrong.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Type of cell division that maintain chromosome number in daughter cell is:

(A) Mitoses
(B) Meiosis
(C) Meiosis-I
(D) Anaphase-I

Q2. In the presence of enzymes, chemical reactions proceed at a:

(A) Faster rate
(B) Slower rate
(C) Constant rate
(D) Medium rate

Q3. If the reactants of chemical reactions are ATP+H2O → ... then the products will be;

(A) AMP+Pi+7.3Kcal
(B) ADP+Pi+7.3Kcal
(C) ATP+Pi+7.3Kcal
(D) ADP+7.3Kcal

Q4. Very intense light is:

(A) Useful for chlorophyll
(B) Increase photosynthesis
(C) Damages chlorophyll
(D) Help chlorophyll formation

Q5. Which of the given vitamin is needed for the growth and repair of body tissues?

(A) Vitamin A
(B) Vitamin B
(C) Vitamin C
(D) Vitamin D

Q6. Lungs are responsible for adding oxygen to the deoxygenated blood that they receive from;

(A) Hepatic artery
(B) Renal artery
(C) Coronary artery
(D) Pulmonary artery

Q7. The study of form, shapes and structure of organisms is;

(A) Morphology
(B) Anatomy
(C) Histology
(D) Physiology

Q8. Collected data is analyzed by using;

(A) Biological method
(B) Statistical method
(C) Mathematical method
(D) Ecological method

Q9. Which of the given is an example of Protista?

(A) Penicillium
(B) Mosses
(C) Smut
(D) Euglena

Q10. Mathias Schleiden claimed that;

(A) Nucleus is spherical body in plant cell
(B) All plants are made up of cells
(C) All living organisms are made of cellular tissues
(D) Every cell comes from a cell

Q11. Which substance is present in higher concentration after the food is digested in small intestine?

(A) Glucose
(B) O2
(C) CO2
(D) Glycogen

Q12. In which of the given phase, neurons cells remain for indefinite period of time?

(A) G 1-Phase
(B) G 2-Phase
(C) G 0-Phase
(D) S-Phase

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 45 minutes.

Q2. Attempt any EIGHT parts. Each part carries FOUR marks.

1) Give correct sequence of biological organization?
2) Write the names of two mammals and two fish species which are over hunted in Pakistan?
3) Briefly explain smooth endoplasmic reticulum?

4) Briefly write the structure and function of fibrous connective tissue?

5) Explain the role of mitosis in growth of organism with an example?

6) Define necrosis. How it causes inflammation?

7) Briefly explain activators. Give TWO examples as well?

8) Define electron transport chain in aerobic respiration.

9) Write the components of saliva and gastric juice.

10) How does the high and low solute concentration in stomata affect the rate of transpiration?

11) Define the "Dubb" sound produced in the heart emphasizing on the opening and closing of different valves.

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 24

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Ratio and proportion are used in solving biological problems. Briefly explain.

(b) Define classification. Write the taxonomic hierarchy of man (human being).

Q4.

(a) What is cytoplasm? Write down its TWO functions.

(b) List THREE events that occur in anaphase-1.

Q5.

(a) Explain induced fit model with the help of diagram.

(b) Explain the role of chlorophyll in photosynthesis.

Q6.

(a) Write causes and symptoms of Anaemia.

(b) Write down the steps of translocation of food in plants.